QUANT

Venn Diagrams

Assignment 02

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Venn Diagram Online Assignment -02

1. In an office there are 250 officers. The dress code at the office requires that if a tie is worn it should be either a pink tie or a red tie. 160 officers wear pink ties and 90 wear red ties. If the number of officers wearing pink ties only is thrice the number of officers wearing red tie only, find the percentage of officers who wear either.

1. 22 % 2. 55 % 3. 15 % 4. 44 %

2. In a village of 1000 population each person reads either the Times of India or the Free Press. 112 persons read only the Times of India, 512 persons read only the Free Press and the remaining persons read both. Find the total number of persons reading the Times of India.

1. 888 2. 488 3. 788 4. 688

3. Find the total number of persons reading the Free Press as per the information given in the above question.

1. 888 2. 488 3. 788 4. 688

4. In a group of 800 people, 550 can speak Hindi and 450 can speak English. How many can speak both Hindi and English?

1. 220 2. 200 3. 260 4. 245

5. In a class of 35 students, 24 like to play cricket and 16 like to play football. Also, each student likes to play at least one of the two games. How many students like to play both cricket and football?

1. 8 2. 4 3. 5 4. 6

6. In a group of 50 people, 35 speak Hindi, 25 speak both English and Hindi and all the people speak at least one of the two languages. How many people speak only English and not Hindi?

1. 28 2. 15 3. 10 4. 20

7. There are 40 students in a Chemistry class and 60 students in a Physics class. Find the number of students which are either in Physics class or Chemistry class if the two classes meet at different hours and 20 students are enrolled in both the subjects.

1. 80 2. 85 3. 90 4. 92

8. In a survey of 700 students in a college, 180 were listed as drinking Limca, 275 as drinking Miranda and 95 were listed as both drinking Limca as well as Miranda. Find how many students were drinking neither Limca nor Miranda.

1. 340 2. 355 3. 330 4. 370

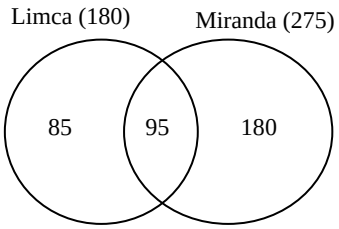
9. A survey shows that 63% of the Americans like cheese whereas 76% like apples. If x% of the Americans like both cheese and apples, find the value of x.

1. $39 \le x \le 65$ 2. $30 \le x \le 60$ 3. $25 \le x \le 50$ 4. $39 \le x \le 63$

10. A market research group conducted a survey of 2000 consumers and reported that 1720 consumers liked product P_1 and 1450 consumers liked product P2. What is the least number that must have liked both the products?

1. 1170 2. 1050 3. 1125 4. 1075

Answer Key & Explanations

Sr. No.	Key	Explanation
1	1	Assume there are x persons in the office who wear either tie. Now the equation will be $160 - x = 3(90 - x)$ $\rightarrow 2x = 110 \rightarrow x = 55$. Now percentage is $100 \times \frac{55}{250} = 22\%$.
2	2	Out of 1,000 who definitely read one of these two newspapers, there are 512 who read only free press. This implies the remaining 1,000 – 512 = 488 read Times of India.
3	1	Similarly 1,000 – 112 = 888.
4	2	Let H denote the set of people speaking Hindi and £ denote the set of people speaking English. We are given that: n (H) = 550, n (E) = 450 and n (H∪E) = 800 and we have to find n(H∩E). Now, n(H∪E) = n(H) + n(E) -n(H∩E) ⇒ n(H∩E) = n(H) + n(E) -n(H∪E) = 550 + 450 - 800 = 200. Hence, 200 persons can speak both Hindi and English.
5	3	Let C be the set of students who like to play cricket and F be the set of students who I like to play football. Then, C∪F is the set of students who like to play at least one game and, C∩F is the set of all students who like to play both games. It is given that n (C) = 24, n (F) =16, n (C∪F) = 35 and we have to find n(C∩F). Now, n(C∪F) = n(C)+n(F)-n(C∩F) ⇒ n(C∩F) = n(C) + n(F) - n(C∪F) = 24 +16 - 35 = 5.
6	2	Let H denote the set of people speaking Hindi and E the set of people speaking English. Then, it is given that: n(H∪E) = 50, n(H) = 35, n(H∩E) = 25. Now, n(E - H) = n(H∪E) -n (H) = 50-35 = 15 Thus the number of people speaking English but not Hindi is 15.
7	1	Let A be the set of students in Chemistry class and B be the set of students in Physics class. It is given that n (A) = 40 and n (B) = 60. We have to find n(A∪B). If two classes meet at different timings then there can be some students attending both the classes. It is given that the number of such students is 20 i.e. n (A∩B) = 20. ∴ n(A∪B) = n(A) + n (B) -n(A∩B) = 40 + 60 - 20 = 80.
8	1	 Hence 85 +95 +180 = 360 students drink atleast one of Limca or Miranda. So the remaining 700 – 360 = 340 students neither drink Limca or Miranda.
9	4	Let A denote the set of Americans who like cheese and let B denote those who like apples. Let the population of America be 100. Then, n(A) = 63, n(B) =76. Now, n(A∩B) = n(A) + n(B) - n(A∪B) ⇒ n(A∩B) = 63 +76 - n(A∪B) = 139 - n(A∪B) But, n(A∪B) ≤ 100. ⇒ - n(A∪B) ≥ -100 ⇒ 139 - n(A∪B) ≥ 139 - 100 ⇒ 139 - n (A∪B) ≥ 39 ⇒ (A∪B) ≥ 39 .....(i) Now, A∪B ⊆ A and A∪B ⊆ B ⇒ A∩B ≤ n(A) and n(A∩B) ≤ n(B) ⇒ n(A∩B) ≤ 63 and n(A∩B) ≤ 76 ⇒ n(A∩B) ≤ 63 ……(ii) From (i) and (ii), we obtain 39 ≤ n(A∩B) ≤ 63 ⇒ 39 ≤ x ≤ 63.
10	1	Let u be the set of all consumers who were questioned, A be the set of consumers who liked product P_1 and B be the set of consumers who liked the product P_2. It is given that n(u) = 2000, n (A) = 1720, n(B) =1450. ∴ n(A∪B) = n(A) + n(B) - n (A∩B) = 1720 + 1450 - n (A∩B) = 3170 - n(A∩B) Now, A∪B ⊂ u ⇒ n(A∪B) ≤ n(u) ⇒ 3170 - n(A∩B) ≤ 2000 ⇒ 3170 - 2000 ≤ n(A∩B) ⇒ n (A∩B) ≥ 1170 Thus, the least value of n (A∩B) is 1170. Hence, the least number of consumer who liked both the products is 1170.